The Lattice Problem for Models of PA

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Substructure / Interstructure Latttices

1. If $\mathcal{M} \models \mathrm{PA}$, then $\mathrm{Lt}(\mathcal{M}) = (\{\mathcal{K} : \mathcal{K} \preccurlyeq \mathcal{M}\}, \preccurlyeq)$ is a lattice, called the substructure lattice of $\mathcal{M}$.
2. If $\mathcal{M} \preccurlyeq \mathcal{N}$, then $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) = (\{\mathcal{K} : \mathcal{M} \preccurlyeq \mathcal{K} \preccurlyeq \mathcal{N}\}, \preccurlyeq)$ is a lattice, called the interstructure lattice between $\mathcal{M}$ and $\mathcal{N}$.

Question

For which finite lattices L is there $\mathcal{M}$ such that $\mathrm{Lt}(\mathcal{M}) \cong L$?

Similarly for interstructure lattices: given (countable / nonstandard) $\mathcal{M}$, for which finite lattices L is there $\mathcal{M} \prec \mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$?

We focus on the interstructure lattice question as it is more general (take $\mathcal{M} = \mathrm{Scl}(0)$).

Some Results

Suppose $\mathcal{M} \models \mathrm{PA}$.

1. If $\mathcal{K} \in \mathrm{Lt}(\mathcal{M})$ is compact, then it is finitely generated (ie, $\mathcal{K} = \mathrm{Scl}(\bar{a})$ for some finite $\bar{a} \in \mathcal{M}$; by coding, can take $\mathcal{K} = \mathrm{Scl}(a)$ for a single $a \in M$).
2. $\mathrm{Lt}(\mathcal{M})$ is complete.
3. Every $\mathcal{K} \in \mathrm{Lt}(\mathcal{M})$ is the supremum of a set of compact elements.
4. Every compact $\mathcal{K} \in \mathrm{Lt}(\mathcal{M})$ has countably many compact predecessors.

Taken together, (2)-(4): $\mathrm{Lt}(\mathcal{M})$ is $\aleph_1$-algebraic. Similarly, if $|M| = \kappa$ and $\mathcal{M} \prec \mathcal{N}$, then $\mathrm{Lt}(\mathcal{N}/\mathcal{M})$ is κ^+-algebraic.

Other restrictions?

Essentially no other restrictions are known about which lattices can be the substructure lattice of a model of PA! In particular, there is nothing that rules out any finite lattice from being such a substructure lattice.

Problem

Is every finite lattice L the substructure lattice of some $\mathcal{M} \models \mathrm{PA}$?

Flavors of results

There are essentially three kinds of results that are known for classes of lattices $\mathcal{L}$:

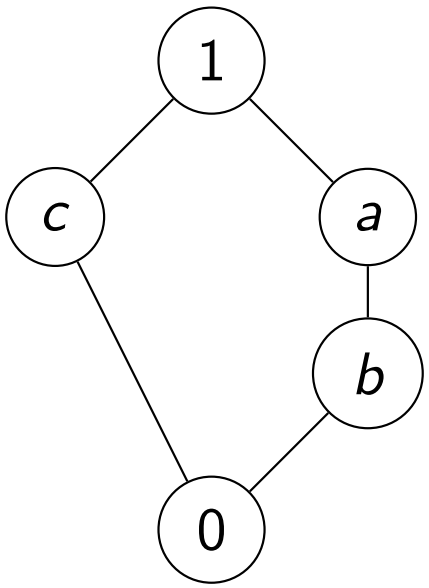
1. For every $L \in \mathcal{L}$ and every $\mathcal{M} \models \mathrm{PA}$, there is $\mathcal{N} \models \mathrm{PA}$ such that $\mathcal{M} \prec \mathcal{N}$ and $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.
2. For every $L \in \mathcal{L}$ and every countable $\mathcal{M} \models \mathrm{PA}$, there is $\mathcal{N} \models \mathrm{PA}$ such that $\mathcal{M} \prec \mathcal{N}$ and $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.
3. For every $L \in \mathcal{L}$ and every countable, nonstandard $\mathcal{M} \models \mathrm{PA}$, there is $\mathcal{N} \models \mathrm{PA}$ such that $\mathcal{M} \prec \mathcal{N}$ and $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.

Examples (1)

- Gaifman (1965, published 1976): If I is any set, then every $\mathcal{M} \models \mathrm{PA}$ has an elementary (end) extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong (\mathcal{P}(I), \subseteq)$.
- Mills (1979): Let D be a distributive lattice. Then D is $\aleph_1$-algebraic if and only if every $\mathcal{M}$ has an elementary (end) extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong D$.

Example (2)

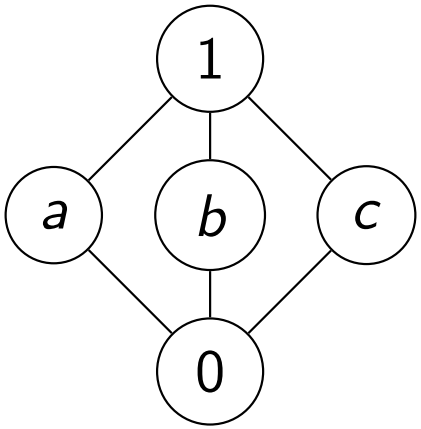
$\mathbf{N}_5$: Pentagon lattice



- Wilkie (1977): for every countable $\mathcal{M} \models \mathrm{PA}$, there is $\mathcal{N}$ such that $\mathcal{M} \prec \mathcal{N}$ and $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{N}_5$.
- Schmerl: there are uncountable $\mathcal{M}$ for which no $\mathcal{N} \succ \mathcal{M}$ exists with $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{N}_5$.

Example (3)

$\mathbf{M}_3$: modular lattice with three atoms



- Paris (1977) / Schmerl (1986): Every countable, nonstandard $\mathcal{M} \models \mathrm{PA}$ has a cofinal elementary extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{M}_3$.
- If $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{M}_3$, then $\mathcal{N}$ must be a cofinal extension of $\mathcal{M}$; in particular, $\mathbb{N}$ does not have an elementary extension $\mathcal{M}$ where $\mathrm{Lt}(\mathcal{M}) \cong \mathbf{M}_3$.

Unknown

- For each n, the lattice $\mathbf{M}_n$ is the lattice with $n+2$ elements: 0, 1, and n atoms in between.
- $\mathbf{M}_1$ is the three element chain ($\mathbf{3}$) and $\mathbf{M}_2$ is the Boolean algebra on a 2-element set $\mathbf{B}_2$: both are distributive (and hence: every $\mathcal{M} \models \mathrm{PA}$ has an elementary extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{3}, \mathbf{B}_2$.
- $\mathbf{M}_n$ for $n \geq 3$ are non-distributive.

Problem (Embarrassing)

Is $\mathbf{M}_{16}$ a substructure lattice?

$n = 16$ is the smallest case for which this is unknown.

Schmerl

- The most general results we have rely on a technique devised by Schmerl (1986): representations of lattices.
- A representation: $\alpha : L \to \mathrm{Eq}(A)$ associates an equivalence relation Θ to each element $r \in L$ (subject to some constraints).
- Idea: given a countable, nonstandard $\mathcal{M}$, describe a type $p(x)$ which ensures that every $\mathcal{M}$-definable function $t(x)$ induces exactly one of those Θ (on a large definable set).

To motivate the definitions / results, let's go the other way, starting with $\mathcal{M} \prec \mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{B}_2$.

The Lattice $\mathbf{B}_2$

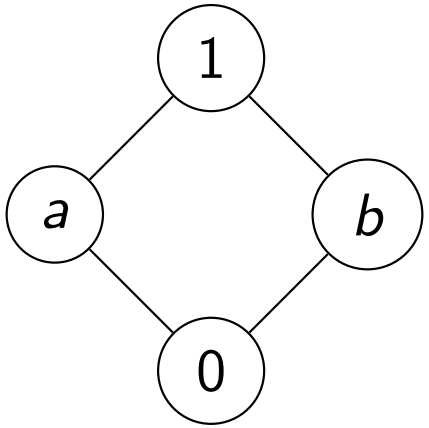


Figure: The lattice $\mathbf{B}_2$

Fix $\mathcal{M} \prec \mathcal{N}$ is such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{B}_2$.

Generators

$\mathbf{B}_2$ is finite: so each element is compact. There are $a, b \in N$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M})$ is exactly:

$$\begin{tikzcd}[column sep=small]
 & \mathcal{M}(\langle a, b\rangle) \arrow[dl, dash] \arrow[dr, dash] & \\
\mathcal{M}(a) \arrow[dr, dash] & & \mathcal{M}(b) \arrow[dl, dash] \\
 & \mathcal{M} &
\end{tikzcd}$$

In fact: $\mathrm{tp}(\langle a, b\rangle/\mathcal{M})$ knows this! (How can we make this precise?)

$\mathrm{tp}(\langle a, b\rangle)$

Suppose $c \in N$. Let t be such that $\mathcal{N} \models t(\langle a, b\rangle) = c$. One of the following must hold:

1. $c \in M$.
2. $\mathcal{M}(c) = \mathcal{M}(a)$.
3. $\mathcal{M}(c) = \mathcal{M}(b)$.
4. $\mathcal{M}(c) = \mathcal{N}$.

In fact, there a formula $\phi \in \mathrm{tp}(\langle a, b\rangle/\mathcal{M})$ ensuring that one of these holds.

Notice

Let π_1 and π_2 be the projection functions:
$\pi_1(\langle x, y\rangle) = x, \pi_2(\langle x, y\rangle) = y$.

Whenever t is an $\mathcal{M}$-definable (total) function, consider the equivalence relation induced by t: $x \sim y$ iff $t(x) = t(y)$. We can find an infinite, $\mathcal{M}$-definable X such that $\langle a, b\rangle \in X^{\mathcal{N}}$ and one of the following holds on X:

1. $\sim$ is trivial (ie, for all $x, y \in X$, $x \sim y$).
2. $x \sim y$ iff $\pi_1(x) = \pi_1(y)$.
3. $x \sim y$ iff $\pi_2(x) = \pi_2(y)$.
4. $\sim$ is discrete (ie, for all $x, y \in X$, $x \sim y$ iff $x = y$).

Definitions

Let X be a set and L a finite lattice. Then:

1. $\mathrm{Eq}(X)$ is the lattice of equivalence relations on X, with top element $\mathbf{1}_X = X \times X$ (trivial relation) and bottom element $\mathbf{0}_X = \{(a, a) : a \in X\}$ (discrete relation).
2. $\alpha : L \to \mathrm{Eq}(X)$ is a representation if it is one to one and:
   - $\alpha(0_L) = \mathbf{1}_X$ ($\alpha(0)$ is trivial),
   - $\alpha(1_L) = \mathbf{0}_X$ ($\alpha(1)$ is discrete), and
   - $\alpha(x \vee y) = \alpha(x) \wedge \alpha(y)$.

That is, a representation picks out specific equivalence relations on X, one for each $r \in L$. Ensure that if $x \leq y$, then $\alpha(y)$ refines $\alpha(x)$.

Example

In any $\mathcal{M} \models \mathrm{PA}$, let X be the set of pairs

$$\{\langle x, y\rangle : x \in M, y \in M, \text{ and } x < y\}.$$

Define $\alpha : \mathbf{B}_2 \to \mathrm{Eq}(X)$:

- $\alpha(0)$ is trivial,
- $(\langle a_1, a_2\rangle, \langle b_1, b_2\rangle) \in \alpha(a)$ iff $a_1 = b_1$,
- $(\langle a_1, a_2\rangle, \langle b_1, b_2\rangle) \in \alpha(b)$ iff $a_2 = b_2$, and
- $\alpha(1)$ is discrete.

(Clearly a representation.)

More definitions

Let $\alpha : L \to \mathrm{Eq}(X)$ be a representation. Then:

1. If $Y \subseteq X$, then $\alpha|Y : L \to \mathrm{Eq}(Y)$ is defined by $(\alpha|Y)(r) = \alpha(r) \cap Y^2$ for each $r \in L$.
2. If $\Theta \in \mathrm{Eq}(X)$, Θ is canonical for α if there is $r \in L$ such that for all $x, y \in X$, $(x, y) \in \Theta$ iff $(x, y) \in \alpha(r)$.

Constructing $\mathbf{B}_2$

Let $\mathcal{M} \models \mathrm{PA}$ be countable. Fix X to be the set of pairs, and $\alpha : \mathbf{B}_2 \to \mathrm{Eq}(X)$ as before. We wish to construct a complete type $p(x)$ such that if c realizes $p(x)$, then $\mathrm{Lt}(\mathcal{M}(c)/\mathcal{M}) \cong \mathbf{B}_2$.

We do this by constructing a sequence $X_0 \supseteq X_1 \supseteq \ldots$ of definable sets that will generate $p(x)$; that is,

$$p(x) = \{\phi(x) : \exists n \in \omega(X_n \subseteq \phi(\mathcal{M}))\}.$$

We ensure that $p(x)$ knows that $\mathrm{Lt}(\mathcal{M}(c)/\mathcal{M}) \cong \mathbf{B}_2$: that is, for each $t(x)$, we ensure that there is X_n where (the equivalence relation induced by) t is canonical for $\alpha|X_n$.

(Skipping details.)

Definitions

Definition (Schmerl 1986)

Let $\alpha : L \to \mathrm{Eq}(X)$ be a representation.

1. α has the 0-*canonical partition property*, or is 0-*CPP*, if for each $r \in L$, $\alpha(r)$ does not have exactly two classes.
2. α is $(n+1)$-*CPP* if, for each $\Theta \in \mathrm{Eq}(X)$ there is $Y \subseteq X$ such that $\alpha|Y$ is an n-CPP representation and $\Theta \cap Y^2$ is canonical for $\alpha|Y$.

The α we used was n-CPP for every $n \in \omega$.

Main Results

Theorem (Schmerl 1986)

Let $\mathcal{M} \models \mathrm{PA}$ be countable and nonstandard and let L be a finite lattice. Then the following are equivalent:

1. *$\mathcal{M}$ has an elementary extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.*
2. *L has an n-CPP representation for each $n \in \omega$.*
3. *$\mathcal{M}$ has a cofinal elementary extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.*

The Lattice Problem

So how do we attack the lattice problem in general?

- For each finite lattice L, try to find a specific representation?
- Is there a proof that all finite lattices have n-CPP representations for every $n \in \omega$? (not yet)
- Is there a proof that all finite lattices in some infinite class have n-CPP representations for every $n \in \omega$? (Yes!)

Congruence Lattices

Let A be a set, I an index set, and for each $i \in I$, $f_i : A^n \to A$ (for some $n \in \omega$). Then $(A, \langle f_i : i \in I \rangle)$ is called an algebra.

If $\mathcal{A} = (A, \langle f_i : i \in I \rangle)$ is an algebra, then $\theta \in \mathrm{Eq}(A)$ is a congruence if it commutes with all f_i.

$\mathrm{Cg}(\mathcal{A})$ is the set of all congruences on $\mathcal{A}$; sublattice of $\mathrm{Eq}(A)$; such lattices are referred to as congruence lattices.

Congruence representations

If L is a lattice, then L^d is the lattice with the ordering reversed.

Definition

Let L be a finite lattice and $\alpha : L \to \mathrm{Eq}(A)$ be a representation. Then α is a congruence representation if there is an algebra $\mathcal{A}$ such that $\alpha \cong \mathrm{Cg}(\mathcal{A})^d$. (Such a representation is a finite congruence representation if $\mathcal{A}$ can be taken to be finite.)

Congruence Representation Theorem

Theorem (Schmerl 1993)

Suppose L is a finite lattice with a finite congruence representation. Then for every countable, nonstandard $\mathcal{M}$, there is $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.

Proof involves nontrivial combinatorics: Prömel-Voigt (canonical Hales-Jewett)! To illustrate its importance, we consider the example of $\mathbf{M}_3$.

(Aside: Abstract: "Some new finite lattices (for example, $\mathbf{M}_4, \mathbf{M}_7$, and the hexagon lattice) are shown to be isomorphic to the lattice of elementary substructures of a model of Peano Arithmetic.")

The lattice $\mathbf{M}_3$

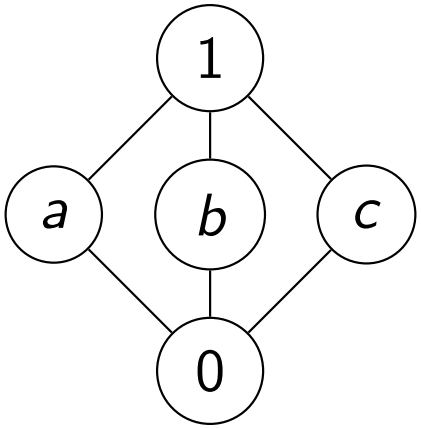


Figure: The lattice $\mathbf{M}_3$.

- $\mathbf{M}_3$ is isomorphic to the lattice of partitions of a three-element set $\{0, 1, 2\}$.
- Call this isomorphism $\alpha : \mathbf{M}_3 \to \mathrm{Eq}(3)$.
- This is a congruence representation: the algebra with no operations (or just the identity).

Powers

Let 3^m refer to the set of m-tuples of elements of $\{0, 1, 2\}$. Define $\alpha^m : \mathbf{M}_3 \to 3^m$ by $(s, t) \in \alpha^m(r)$ iff $(s_i, t_i) \in \alpha(r)$ for each $i < m$. Then:

Lemma

For every m, there is n such that whenever $\Theta \in \mathrm{Eq}(3^n)$, there is "$m$-dimensional" $A \subseteq 3^n$ and $r \in \mathbf{M}_3$ such that whenever $\bar{x}, \bar{y} \in A$, $(\bar{x}, \bar{y}) \in \Theta$ iff $(x_i, y_i) \in \alpha(r)$ for each $i < n$.

- m-dimensional? "There are m pairs (x_i, y_i) to check."
- Ex: $\{(a, b, 1, a, 0) : a, b \in 3\}$ is a 2-dimensional subset of 3^5.
- **Notation**: the above tells us $\alpha^n \to \alpha^m$: ie, each equivalence relation can look like $\alpha^n(r)$ for some r on an m-dimensional subset.
- How do we prove this? Previously known result in combinatorics: Prömel-Voigt / canonical Hales-Jewett.

HJ, PV

Theorem (Hales-Jewett)

For every finite set A and every m, there is n such that every finite coloring A^n contains a monochromatic m-dimensional combinatorial subset.

There is a "canonical" version of this (akin to the "canonical" version of Ramsey's Theorem).

Theorem (Prömel-Voigt)

Let A be a finite set. For every m, there is n such that whenever $\Theta \in \mathrm{Eq}(A^n)$, there is an m-dimensional $X \subseteq A^n$ and $\theta \in \mathrm{Eq}(A)$ such that whenever $\bar{x}, \bar{y} \in X$, $(\bar{x}, \bar{y}) \in \Theta$ iff $(x_i, y_i) \in \theta$ for each $i < n$.

Look closely: for $A = \{0, 1, 2\}$, this is exactly what the Lemma on the previous slide said.

Technique

Start with $\alpha_0 = \alpha^2$. Then for each n, we can find chains $\alpha_n \to \alpha_{n-1} \to \ldots \to \alpha_0$. Overspill: $\alpha_c : \mathbf{M}_3 \to \mathrm{Eq}(3^d)$ for some nonstandard c, d. The rest is a standard technique:

- Enumerate Skolem functions $t_0, t_1, \ldots$
- Start with $A_0 = 3^d$
- Given A_i, t_i, find $r \in \mathbf{M}_3$ and "large" $A_{i+1} \subseteq A_i$ such that whenever $\bar{x}, \bar{y} \in A$, $t_i(\bar{x}) = t_i(\bar{y})$ iff $(x_j, y_j) \in \alpha(r)$ for all $j < c$.
- Now the sets $A_0 \supseteq A_1 \supseteq \ldots$ generate a type $p(x) = \{\phi(x) : \phi(M) \supseteq A_i \text{ for some } i \in \omega\}$.
- If a realizes $p(x)$, then $\mathrm{Lt}(\mathcal{M}(a)/\mathcal{M}) \cong \mathbf{M}_3$.

Known and unknown

- Known: every algebraic lattice is isomorphic to a congruence algebra.
- Open ("finite lattice representation problem"): is every finite lattice isomorphic to $\mathrm{Cg}(\mathcal{A})$ for some finite $\mathcal{A}$?
- Every finite lattice that is currently known to be a substructure lattice has a congruence representation.
- Unknown: is there a finite substructure lattice that is not a congruence representation?

Thank you!

Recently surveyed the lattice problem with Roman Kossak, "The Lattice Problem for models of PA" (BSL).

Some other references:

- Kossak and Schmerl, The Structure of Models of Peano Arithmetic (Chapter 4)
- Schmerl, Substructure lattices of models of Peano arithmetic. Logic Colloquium 84, pp. 225-243 (1986).
- Schmerl, The pentagon as a substructure lattice of models of Peano arithmetic. The Journal of Symbolic Logic, pp. 1-25 (2024).